ABN SEAL COLLEGE

COOCHBEHAR

PANCHANAN BARMA UNIVERSITY



Biogeochemical Cycle

Carbon • Nitrogen • Phosphorus

B.Sc. 2nd Semester | CBPBU

Created by Bikram Singha

Scanned with OKEN Scanner

∅ Biogeochemical Cycle

Biogeochemical cycle

- Biological Components → Plants, Animals & Microorganisms
- Geological Components → soil, rock, air, water
- chemical process → reaction that transform substrate
- circulation b/w three component or process

→ A biogeochemical cycle is the natural process by which chemical elements (carbon, nitrogen, oxygen, sulphur, phosphorus) circulate between living organisms & the non living environment through chemical changes.
→ example- Carbon cycle, Nitrogen cycle, phosphorus cycle

Nitrogen cycle

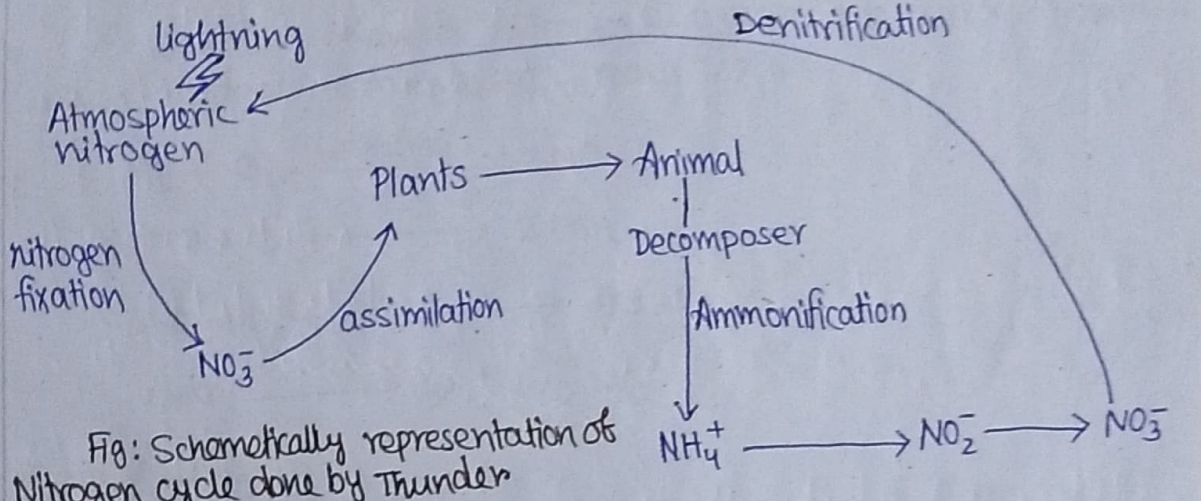


Fig: Schematically representation of Nitrogen cycle done by Thunder

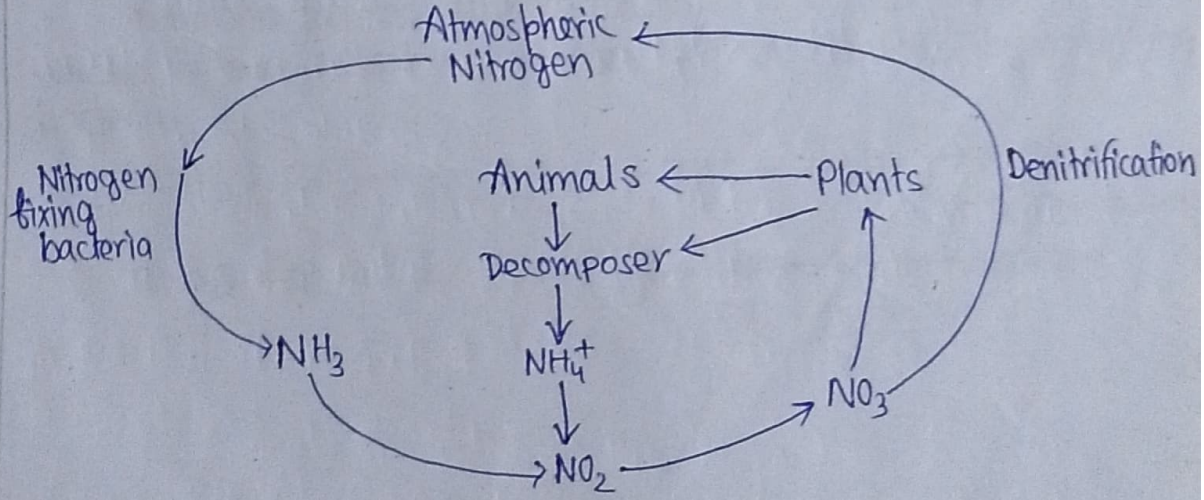


Fig: Schematically representation of N_2 cycle done by bacteria.

Nitrogen cycle-

The nitrogen cycle is the cyclic movement of Nitrogen b/w atmosphere, soil & living organisms. It maintains the balance of Nitrogen in nature.

The ~~set~~ steps of N_2 cycle as follows-

① Nitrogen fixation-

Atmospheric nitrogen (N_2) cannot be directly used by most plants. ~~It is cant~~ Thus, it is converted into usable components like ammonia, Nitrate etc.

—By Bacteria

Nitrogen-fixing bacteria convert atmospheric nitrogen into ammonia

Rhizobium Azotobacter

$$N_2 + 8H^+ + 8e^- \longrightarrow 2NH_3 + H_2$$

—By lightning

Lightning converts Nitrogen & oxygen into nitrogen oxides which dissolve in rainwater & reach the soil.

$$N_2 + O_2 \xrightarrow{\text{lightning}} 2NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

$$4NO_2 + O_2 + 2H_2O \longrightarrow 4HNO_3$$

$$4HNO_3 \longrightarrow 4H^+ + \tfrac{1}{4}NO_3^-$$

② Nitrification-

Ammonia formed during fixation is converted into ~~nitrates & then nitrat ni~~ nitrites & then nitrates by nitrifying bacteria.

$$2NH_3 + 3O_2 \longrightarrow 2NO_2^- + 2H^+ + 2H_2O$$ } → done by nitrosomonas

$$2NO_2^- + O_2 \longrightarrow 2NO_3^-$$ } → ~~done by Nitrobacter~~ eg, Nitrobactor sp.

Scanned with OKEN Scanner

③ Assimilation-

Plant absorb nitrates from the soil through roots & convert them into proteins & other nitrogenous compounds.

Animals obtain nitrogen by eating plants or other animals. $NO_3^- \longrightarrow$ organic nitrogen compounds

④ Ammonification

When plants & animals die, decomposers like bacteria & fungi breakdown proteins & other nitrogenous wastes into ammonia

organic nitrogen $\longrightarrow NH_3$

⑤ Denitrification-

Denitrifying bacteria convert nitrates back into atmospheric nitrogen, completing the cycle

$$2NO_3^- + 4H^+ + 2e^- \longrightarrow 2NO_2^- + 2H_2O$$

$$2NO_2^- + 4H^+ + 2e^- \longrightarrow 2NO + 2H_2O$$

$$2NO + 2H^+ + 2e^- \longrightarrow N_2O + H_2O$$

$$N_2O + 2H^+ + 2e^- \longrightarrow N_2 + H_2O$$

↳ Back to atmosphere

Phosphorus Cycle

The phosphorus cycle is a biogeochemical cycle that describes the movement of phosphorus through the lithosphere hydrosphere & biosphere.

Unlike nitrogen & Carbon cycles, phosphorus has no major gaseous phase because phosphorus mainly occurs as solid phosphate minerals.

Phosphate is essential for-

> DNA & RNA

> ATP (energy transfer)

> ~~Ph~~ Plasma membrane (Phospholipids)

> Bones & teeth

> Plant growth.

Scanned with OKEN Scanner

Main reservoir of ~~Phosphate~~ Phosphorus
The largest ~~se~~ reservoir is-

> Phosphate rock
> Marine sediments.
> Soil minerals.

Main form- Phosphate ion PO_4^{3-}

Steps of the Phosphorus cycle-

1) Weathering of Phosphate Rocks-

Phosphate containing rocks are broken down by-

> Rain
> Wind
> Temperature changes
> microbial action.

This release phosphate ions into soil & water.

$$Ca_3(PO_4)_2 \longrightarrow 3Ca^{2+} + 2PO_4^{3-}$$

2) Absorption / Assimilation by Plants-

Plant absorb dissolved phosphate ions from soil through roots.

$$HPO_4^{2-} + H_2PO_4^{-} \longrightarrow \text{Plant Organic } \text{~~Phou~~ Phosphorus compounds.}$$

Plants Convert phosphate into-

> ~~Nuc~~ nucleic acid
> ATP
> Phospholipids.

3) Transfer through Food Chain

Animals obtain phosphorus by eating plants or other ~~anim~~ animals.

example →

Plant phosphorus → Herbivores → Carnivores.

Scanned with OKEN Scanner

Phosphorus becomes part of
> Bones
> Teeth
> tissues
> ATP

example compound-

Hydroxyapatite in Bones → $Ca_{10}(PO_4)_6(OH)_2$

4) Death & Excretion-

~~Dead~~ Dead organisms & animals wastes return phosphorus to soil & water.

example

urine, dead leaves, dead animals.

5) Decomposition & Mineralization-

Decomposers such as bacteria & fungi convert organic phosphorus back into inorganic phosphate

Organic phosphorus $\xrightarrow{\text{decomposers}}$ PO_4^{3-}

> Recycle phosphorus
> Restores soil fertility

6) Runoff into water bodies-

some phosphate is washed into rivers, lakes & oceans. through rainwater & soil erosion.

* Aquatic plants absorb phosphate &
Excess phosphate can cause
> Eutrophication & Algal Bloom

Scanned with OKEN Scanner

⑦ Sedimentation–

In aquatic system phosphate settles at the bottom & form sediments.

$$PO_4^{3-} + Ca^{2+} \longrightarrow Ca_3(PO_4)_2$$

⑧ Geological uplift–

Tectonic activities uplift marine sediments & phosphate rock to land.

Those rocks again undergo weathering & the cycle repeats.

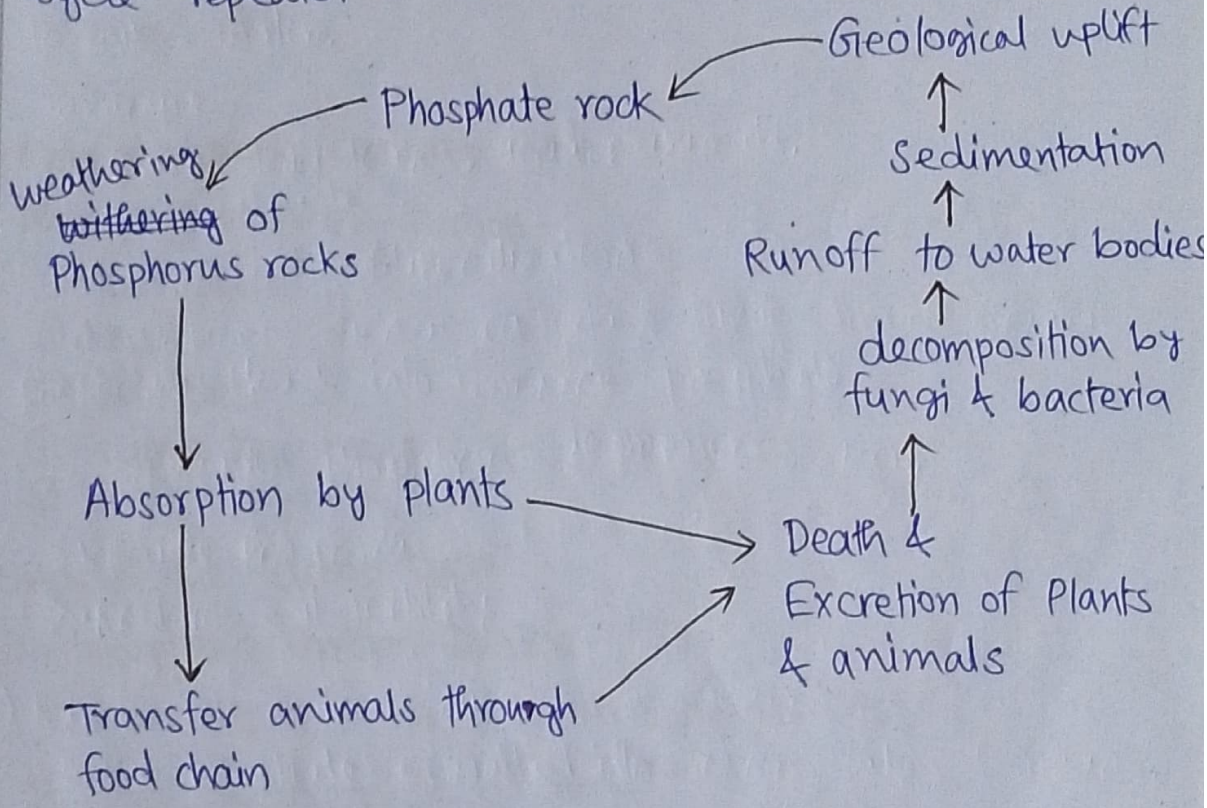


Fig: Schemetic cycle of Phosphorus cycle.

Scanned with OKEN Scanner

Carbon cycle

The Carbon cycle is the biogeochemical cycle through which carbon moves between the- atmosphere, biosphere, hydrosphere lithosphere

Carbon is continuously exchanged among living organisms & the environment.

Importance of Carbon-
- > Carbohydrates formation
- > Protein
- > Fat
- > DNA & RNA
- > ATP

Main atmospheric carbon- CO_2

Main resorvoirs of Carbon-
1. Atmosphere
2. Ocean
3. Fossil fuels
4. Sedimentary rocks
5. Living organisms.

Steps of the Carbon cycle

① Atmospheric Carbon Dioxide-

Carbon exists in atmosphere mainly as- CO_2

It enters the atmosphere from-
- > respiration.
- > combustion.
- > decomposition.
- > volcanic eruption.

② Photosynthesis-

Green plants absorb atmospheric carbon dioxide & convert it into glucose

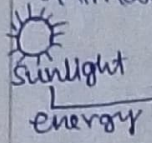
$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

- > Main entry point of carbon into living organisms.
- > Produces food & Oxygen.

Scanned with OKEN Scanner

③ Transfer Through Food Chain–

Animals obtain carbon by eating plants & other animals.



→ Plants ⟶ Herbivores ⟶ Carnivores

Carbon become part of

> proteins
> fats
> Carbohydrates

④ Respiration –

Plants & animals release carbon dioxide back to atmosphere during respiration.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy}$$

→ Return carbon to atmosphere.
→ Release energy.

⑤ Decomposition –

Dead organisms are decomposed by bacteria & fungi. organic carbon is converted into –

> Carbon dioxide
> Methane.

Aerobic decomposition –

$$\text{Organic matter} + O_2 \longrightarrow CO_2 + H_2O$$

Anaerobic decomposition –

$$\text{Organic matter} \longrightarrow CH_4 + CO_2$$

⑥ Formation of fossil fuels –

Some dead organisms buried under pressure for millions of years form –

> Coal
> Petroleum
> natural gas.

Scanned with OKEN Scanner

⑦ Combustion –

Burning fossils fuel & wood releases carbon dioxide

$$C + O_2 \longrightarrow CO_2$$

example-

vehicle fuel buring
Factory emissions
Forest fires.

⑧ Oceanic absorption-

oceans absorb atmospheric carbon dioxide

$$CO_2 + H_2O \longrightarrow H_2CO_3 \text{ (carbonic acid)}$$

$$H_2CO_3 \longrightarrow H^+ + HCO_3^-$$

marine organism use this carbonate to form shells.

⑨ Sedimentation & Rock Formation-

Dead organisms form limestone & carbonate rocks

$$Ca^{2+} + CO_3^{2-} \longrightarrow CaCO_3 \text{ (calcium carbonate)}$$

These rocks store carbon for million of years.

⑩ ~~Vor~~ Volcanic Activity-

Volcanoes release stored carbon dioxide back into atmosphere.

Scanned with OKEN Scanner

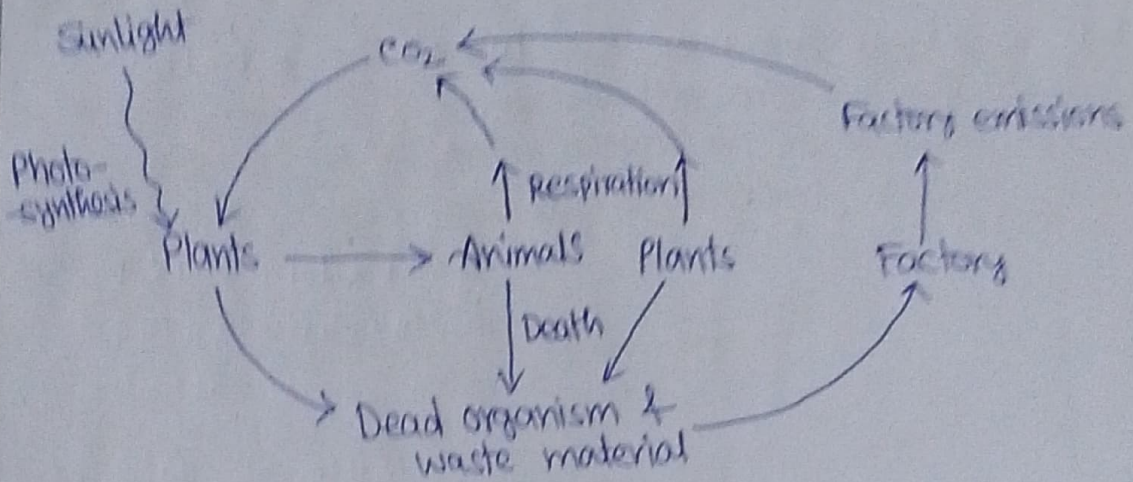


Fig: Cyclic representation of carbon cycle